

GURNIT CHAUHAN

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EDUCATION

Rutgers Business School, Rutgers University

M.S, Information Technology & Analytics (Concentration in Artificial Intelligence, Machine Learning and Data Science)

Newark, NJ

May 2025

• GPA: 3.7

Rutgers Business School, Rutgers University

B.S, Business Analytics and Information Technology, Minor in Data Science

New Brunswick, NJ

May 2024

• GPA: 3.5

• **Relevant Coursework:** Business Forecasting, Data Mining, Project Management, Business Data Management, Applied Statistics, Learning, Analytics for BI, AI Methods with Neural Networks and Deep Learning, Large-Scale Business Data Analysis

PROFESSIONAL EXPERIENCE

Cognizant

Generative AI Extern

Remote

June 2024 - July 2024

- Project 1: Image Classification with Pre-trained Model
 - Developed a classification pipeline for a citywide dog show registration system using a pre-trained CNN model, evaluating AlexNet, VGG, and ResNet to determine the most efficient and accurate architecture, ultimately concluding that VGG performed best.
 - Implemented timing functions, ensured accurate labeling of pet images as "dogs" or "not dogs," and optimized output verification and formatting for streamlined processing.
 - Fine-tuned the VGG model's hyperparameters and preprocessing steps to significantly improve classification accuracy, raising the quality of results from **60%** to over **90%**
- Project 2: Foundation Model Preparation and Fine-Tuning
 - Developed and fine-tuned a PEFT (Parameter-Efficient Fine-Tuning) model by loading and preprocessing datasets, evaluating a pre-trained Hugging Face model, and conducting inference.
 - Saved the fine-tuned model and performed a comprehensive performance evaluation
- Completed the Udacity course and hands-on labs as part of the Externship upskilling program, focusing on Python for AI, Generative AI fundamentals, and applied AI techniques

PSEG

Social Media/Digital Analytics Intern (Extension)

Newark, NJ

Sep 2023 – May 2024

- Utilized Salesforce Analytics Service Cloud and Google Analytics to derive actionable insights, directly driving a **10%** improvement in customer satisfaction through enhanced metrics analysis.
- Led social media optimization efforts, implementing tagging and user management strategies that resulted in a measurable increase in engagement across platforms.

Social Media/Digital Analytics Intern

June 2023 – Aug 2023

- Automated reporting processes by developing VBA macros, improving efficiency by saving **5** hours per week and significantly boosting reporting accuracy.
- Conducted comprehensive mobile app and web testing using SAP and Outage Management systems, ensuring precise identification and resolution of critical issues to maintain operational excellence.

Orion Innovation

Data Science/Analytics Intern

Edison, NJ

July 2022 – August 2022

- Created an AI model by using **Jupyter Notebook** and **Python** libraries after analyzing over **4000** entries of ticket resolution data to find issues, predicting what issues may come in the next **12** months, from the historical model
- Constructed detailed visuals using **Power BI** to measure productivity of each ticket support engineer, given the effects of **10+** variables
- Presented multivariable-backed solutions, resulting in a **60%** increase in workplace productivity as well as more effective and efficient ticket resolution procedures in regard to skill level and fulfilling specific Key Performance Indicators

Rutgers Student Centers & Activities

Building Operations Manager

Piscataway, NJ

May 2021 – June 2023

- Coordinated optimal resource utilization and time management to effectively organize events with a **90%** success rate while adhering to the Student Center's policies and regulations
 - Improved overall initiative and productivity while leading a line staff of **2-4** members by incorporating discipline over process; resulted in **20%** fewer disruptions and complaints
 - Proactively recommended innovative solutions for efficient allocation of Student Center resources; implementation led to successful inventory management and minimized conflicts of conference rooms
 - Developed profitable client relations by solving diverse problems to enrich the guest experience while fostering an inclusive and proactive work environment
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CERTIFICATIONS

- Microsoft Certified: Power BI Data Analyst Associate — Issued [March 2025]
- Microsoft Certified: Azure AI Fundamentals — Issued [August 2025]
- Cognizant Generative AI Externship — Issued [July 2024]

ACADEMIC PROJECTS

Lyric Similarity Generator & Network Graph Visualizer

- Built an ETL pipeline in R to scrape and preprocess thousands of song lyrics from Genius.com. Applied techniques such as normalization, stop-word removal, tokenization, and storage in a structured DataFrame for training purposes
- Trained a Word2Vec model using the CBOW architecture to produce word embeddings. Leveraged cosine similarity to identify contextually similar words, aiding lyricists in finding suitable word alternatives
- Created an interactive R-Shiny dashboard, allowing users to input words and view similar terms through dynamic visualizations, including histograms and a ggplot-based chart
- Implemented a Word Network Graph feature, displaying relationships among contextually similar words with adjustable edge thickness to depict connection strength. This feature was implemented using ggraph, enhancing the user experience by allowing easy exploration of lyrical connections

Anomaly Detection in Credit Card Transactions

- Applied machine learning techniques—including PCA, Isolation Forest, GMM, and Random Forest—to detect fraudulent transactions in a highly imbalanced credit card dataset
- Implemented PCA-based anomaly detection and optimized the reconstruction error threshold, boosting F1-score from 0.06 to 0.19, while maintaining 65% recall for fraud detection and improving interpretability.
- Balanced the dataset using SMOTE and performed hyper-parameter tuning on Isolation Forest and Random Forest models to address class imbalance and enhance detection accuracy.
- Achieved highest performance using Random Forest with class weighting, reaching 96% precision, 74% recall, and a ROC-AUC score of 0.953, outperforming unsupervised models.

Custom GAN Development for Image Synthesis

- Implemented a Vanilla GAN from scratch in TensorFlow, designing a generator that maps 100-dimensional noise to $256 \times 256 \times 3$ images using a Dense-to-Conv2DTranspose stack, and a convolutional discriminator; trained both with Adam ($\text{lr} = 0.0002$, $\beta_1 = 0.5$) and binary cross-entropy adversarial loss.
- Engineered a shard-based data pipeline that normalized 62.3 K celebrity-face images, split them into 20 Apache Parquet files, and streamed batches via tf.data, avoiding GPU/CPU out-of-memory errors and cutting I/O time roughly 3× compared with raw PNG loading.
- Trained for 200 epochs with checkpointing and periodic sample visualization, establishing the team's initial performance baseline with a Fréchet Inception Distance (FID) of 452 on a 5 K held-out celebrity-face set.
- Diagnosed and mitigated common Vanilla-GAN pitfalls—loss oscillation, mode collapse, black-image failure modes—through shard-level shuffling, optimizer tuning, and early detection via saved image grids and logs, informing later architectural choices for DCGAN and StyleGAN.

SKILLS/ACTIVITIES & SPECIAL INTERESTS

- **Technical Skills:** Python, Microsoft Power BI, SQL, Google Analytics, Salesforce, Microsoft Office Suite, R, Tableau, Microsoft Excel
- **Spoken Languages:** Fluent in English and Hindi, Limited Working Proficiency in Spanish
- **Hobbies/Interests:** Fitness, spending time with my golden, Loyal Patriots fan